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Module 5 Introduction

Lesson 1: Parameterized Priors and the Concept of Exchangeability

Lesson 2: Analysis and Applications of Hierarchical Models

Lesson 3: Computational Techniques and Model Validation

Module 5 Summative Assessment

Graded Assignment: Module 5 Summative Assessment
Submitted

Module 5 Summary

Your grade: 86.66%

Your latest: 86.66% • Your highest: 86.66% • To pass you need at least 80%. We keep your highest score.

Next item

Module 5 Summative Assessment

Review Learning Objectives

Instructions

coach

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Assignment details

Due Dec 1, 11:59 PM IST

Attempts 0 left (1 attempts every 8 hours)

Retry

Submitted
Grading in progress

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To pass you need at least 80%. We keep your highest score.

86.66%

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1. What are potential interpretations of hyperparameters? **1 / 1 point**

- Variables representing a hidden force influencing differences in parameter estimates for each experiment
- Variables capturing the underlying relationship between groups
- Neither is true
- (•) Both are true

Correct
Correct! Both interpretations are true and provide valuable insights into the role and meaning of hyperparameters in the context of hierarchical models.

2. Why is it important to check the integrability of the posterior distribution when using improper priors? **1 / 1 point**

- To ensure computational efficiency.
- To avoid overfitting the data.
- To simplify the prior distribution and make simulation possible.
- (•) To verify that the resulting posterior is proper.

Correct
Correct! It is crucial to ensure the resulting posterior distribution is proper.

3. How many types of posterior predictive distributions are there? **1 / 1 point**

- 4
- (•) 2
- 1
- 3

Correct
Correct! We can generate new points for the existing experiments or generate new experiments.

4. Why do we compute pair plots? **1 / 1 point**

- To calculate the means and variances of variables along the diagonal elements.
- To perform statistical hypothesis tests.
- To ensure that the data is normally distributed.
- (•) To visually inspect for strong linear patterns.

Correct
Correct! Pair plots allow us to visually inspect the relationships between all pairs of variables, helping to identify patterns, correlations, divergent transitions and potential outliers.

5. What are divergent transitions? **1 / 1 point**

- Transitions that ensure the sampler has explored the entire parameter space.
- (•) Transitions where sampling has errors due to the posterior distribution's geometry.
- Transitions that show the acceptance rate of the sampler is high and approach infinity asymptotically.
- Transitions that indicate the Markov chain has converged to the target distribution.

Correct
Correct! Divergent transitions occur when the Hamiltonian Monte Carlo (HMC) sampler struggles to accurately follow the geometry of the posterior distribution, often due to regions of high curvature or poor parameter scaling.

6. In the rat tumor example, what distribution is assumed for the parameter θ? **1 / 1 point**

- Normal distribution
- Uniform distribution
- (•) Beta distribution
- Exponential distribution

Correct
The parameter θ is assumed to follow a Beta distribution.

7. What is a key feature of hierarchical models? **1 / 1 point**

- (•) They model data with hyperparameters.
- They always use non-informative priors.
- They assume all data points are independent.
- They are only applicable to large datasets.

Correct
Hierarchical models involve modeling data conditionally on certain parameters, which themselves have a probabilistic specification in terms of hyperparameters.

8. Which of these answers is a potential first step in Bayesian analysis of conjugate hierarchical models? **0 / 1 point**

- Write the joint posterior density in unnormalized form.
- Calculate the marginal likelihood.
- (•) Choose a non-informative hyperprior.
- Simulating the predictive posterior distribution within the existing experimental data.

Incorrect
Choosing a prior is not the first step in the analysis.

9. Why are simple nonhierarchical models usually inappropriate for hierarchical data? **1 / 1 point**

- They assume perfect data.
- They are too complex for small datasets.
- (•) They tend to overfit or underfit data.
- They do not require prior distributions.

Correct
Nonhierarchical models can either overfit or underfit data because they cannot appropriately balance the complexity of the model with the amount of data.

10. What is a major disadvantage of using a point estimate for and in the Beta distribution? **1 / 1 point**

- It requires a large sample size.
- It is computationally intensive.
- (•) It can ignore some posterior uncertainty.
- It always leads to overfitting.

Correct
Using a point estimate for and ignores some posterior uncertainty.

11. What is the posterior predictive distribution used for in hierarchical models? **1 / 1 point**

- (•) To predict future observations
- None of the above
- To update prior distributions
- To update posterior distributions

Correct
Correct! The posterior predictive distribution is used to predict future observations.

12. What is the value of a contour plot of the posterior distribution? **1 / 1 point**

- It identifies the best prior distribution to use.
- It simplifies the calculation of posterior means.
- (•) It provides a visual representation of the density of the posterior distribution.
- It ensures the posterior distribution is normal.

Correct
Correct! A contour plot of the posterior distribution provides a visual representation of the density, showing regions of high and low probability.

13. In the rat tumor example, what is used to simulate the posterior distribution? **0 / 1 point**

- Gamma distribution
- (•) Normal distribution
- Uniform distribution
- Beta distribution

Incorrect
Feedback: Incorrect. A normal distribution is not used in this example.

14. True or False: Predictive posterior distributions of hierarchical models rely only on parameters. **1 / 1 point**

- (•) False
- True

Correct
Correct! Predictive posterior distributions in hierarchical models rely on parameters, hyperparameters, and the overall hierarchical structure, including priors and hyperpriors.

15. What is the primary objection to exchangeable models? **1 / 1 point**

- They are too complex to be understood by most practitioners.
- (•) They assume that all units are identical, which may not reflect reality.
- They require very large datasets to produce reliable results.
- They require the use of hierarchical models and hyperpriors are purely theoretical

Correct
Correct! One of the primary objections to exchangeable models is that they assume all units are identical, which may not accurately reflect the true variability and differences between units in many real-world situations.